

RF Admittance level switch Disc type For Solid Application Model: RFLS-300SD

PRODUCT DESCRIPTION:



The RFLS-300SD RF Admittance type level switch is a simple, low cost, rugged point Level Switch designed for Level detection of Bulk materials in a bins or silo. This level switch can be used for detecting high, low level or for differential level detection and is used in various level detection applications like. cement industries, material handling plants, chemical industry, power utilities industry, food industry, pulp/paper industry, pharmaceuticals industry, water/ treatment plants, foundry industry etc.

The specialty of the probe is that it tends to ignore the build-up on the probe, thus making it suitable for application, which tends to stick on the probe.

the unit shown above is integral Type Disc Probe along with time delay feature, LED indication, fail safe mode etc.

Unit Shown Below is Remote type Disc Probe along with time delay feature, LED indication, fail safe mode etc.

SALIENT FEATURES:

- Ø Easy installation and maintenance.
- Ø Wide range of industrial applications.
- Ø Weatherproof / Flameproof enclosure as per requirement.
- Ø Sturdy Disc Type electrode designed to withstand impact due to accidental clinker formation.
- Ø UHMWPE insulated electrode suitable for operating temperatures up to 80°C.
- Ø PTFE insulated electrode suitable for operating temperatures up to 200°C.
- Ø No moving parts, ensuring no mechanical wear and tear.
- Ø Adjustable differential time delay with ON/OFF timing.
- Ø Fail-Safe High/Low selection feature – field configurable.
- Ø Disc Type probe suitable for bulk solid level detection.
- Ø Suitable for high-level and low-level point level detection.
- Ø Reliable level detection in dusty and demanding industrial environments.
- Ø Available in Integral Type and Remote Type electronics.
- Ø Rugged and heavy-duty construction for industrial applications.
- Ø Customised probe design available as per application requirements.



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OPERATING PRINCIPLE:

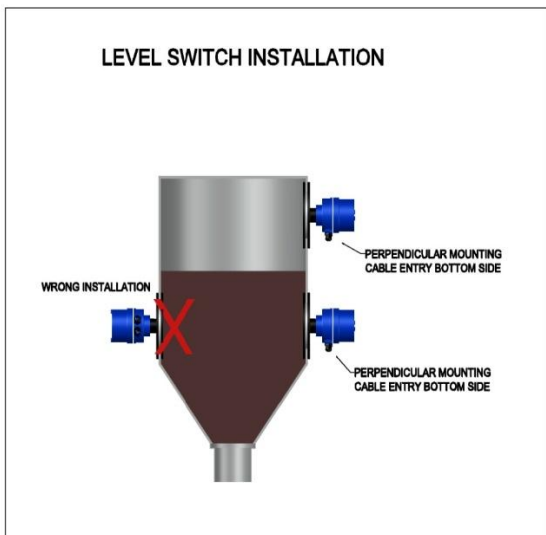
The RFLS-300SD RF Admittance Level Switch operates on the principle of Radio Frequency (RF) Admittance Measurement. The Disc Type Probe and vessel wall form an electrical sensing system. The electronic circuit continuously applies an RF signal to the probe and monitors the electrical admittance between the probe and vessel ground.

When the solid material reaches the Disc Type Sensing Probe, the dielectric properties around the probe change, resulting in a change in RF admittance. The instrument detects this change and, when it exceeds the preset sensitivity level, activates the output relay to provide a reliable High-Level or Low-Level indication.

The Disc Type Probe is designed for solid applications and is suitable for silos, hoppers, bins and storage vessels. Build-Up Compensation / Coating Rejection minimises the effect of material accumulated on the probe, ensuring reliable level detection.

Fail-Safe High/Low Selection, adjustable sensitivity and ON/OFF time delay allow configuration according to different material and process requirements. With no moving parts, the instrument provides reliable long-term operation with minimum mechanical wear and maintenance.

PROBE CONSTRUCTION SUITABLE FOR MATERIAL BUILD-UP:



The COAT Guard electrode uses advanced RF build-up sensing technology to compensate for material coating on the Disc Type Probe. It minimises the effect of accumulated material and prevents false switching.

When the actual material level reaches the sensing probe, the RF admittance changes beyond the compensated condition, and the electronic circuit switches the output to indicate the actual level.

Flange Type mounting with COAT Guard construction provides reliable sensing for sticky, conductive, powdery, granular and coating-forming materials.

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COMPONENTS OF COMPACT LEVEL SWITCH ELECTRICAL CONNECTION ARE EXPLAINED IN FIGURE UNIVERSAL /230 VAC/110VAC/24 VDC

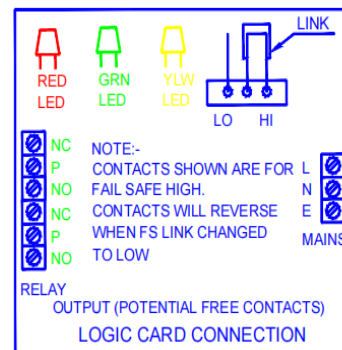
Integral Type



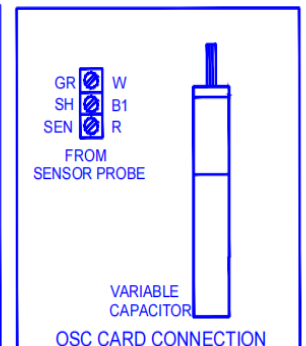
Remote Type



Remote Probe/Controller /RF Cable



Controller Card



Oscillator Card

DATA SHEET

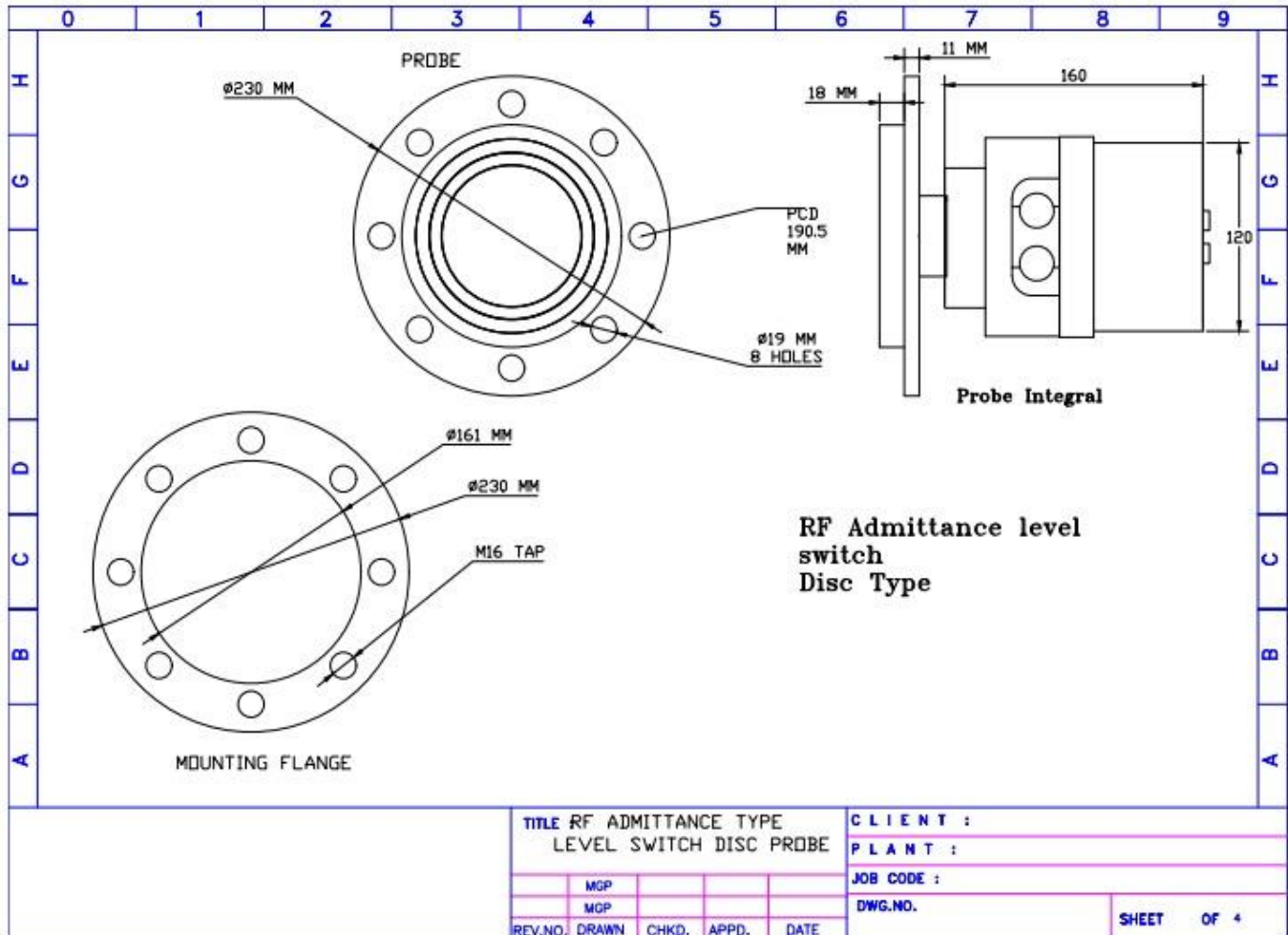
RFLS-300– RF Admittance Level Switch
Selection / Ordering Information

Selection	Code	Description
Model No.	RFLS-300	RF Admittance Level Switch
Application	SD	Solids Disc Type
Mounting Type	IN/RE	Integral / Remote
Insulation	HD/PT	HDPE / PTFE
Probe Material	04 /16/MS/ SM	SS304/ SS316 /Mild steel/ Special material
Power Supply	AC/DC/A1/UN	230VAC / 24VDC / 110VAC / Universal
Output	DP	DPDT Relay
Switching Delay	ND / DA	No Delay / Adjustable
Indication	L	LED
Enclosure	WP / FL	Weatherproof / Flameproof
Connection	F	Flanged
Flange Rating	A1 / CS	ANSI 150# /Customer special Requirement
MOC-Process Connection	S4 / S16/MS/SL	SS304 / SS316 /Mild steel / Special material
Flange Size	04/PS	4"/ Special Process
Pressure	P1 /P2/P4	Standard 0 to 2KG/ 2 To 10kg / Customised

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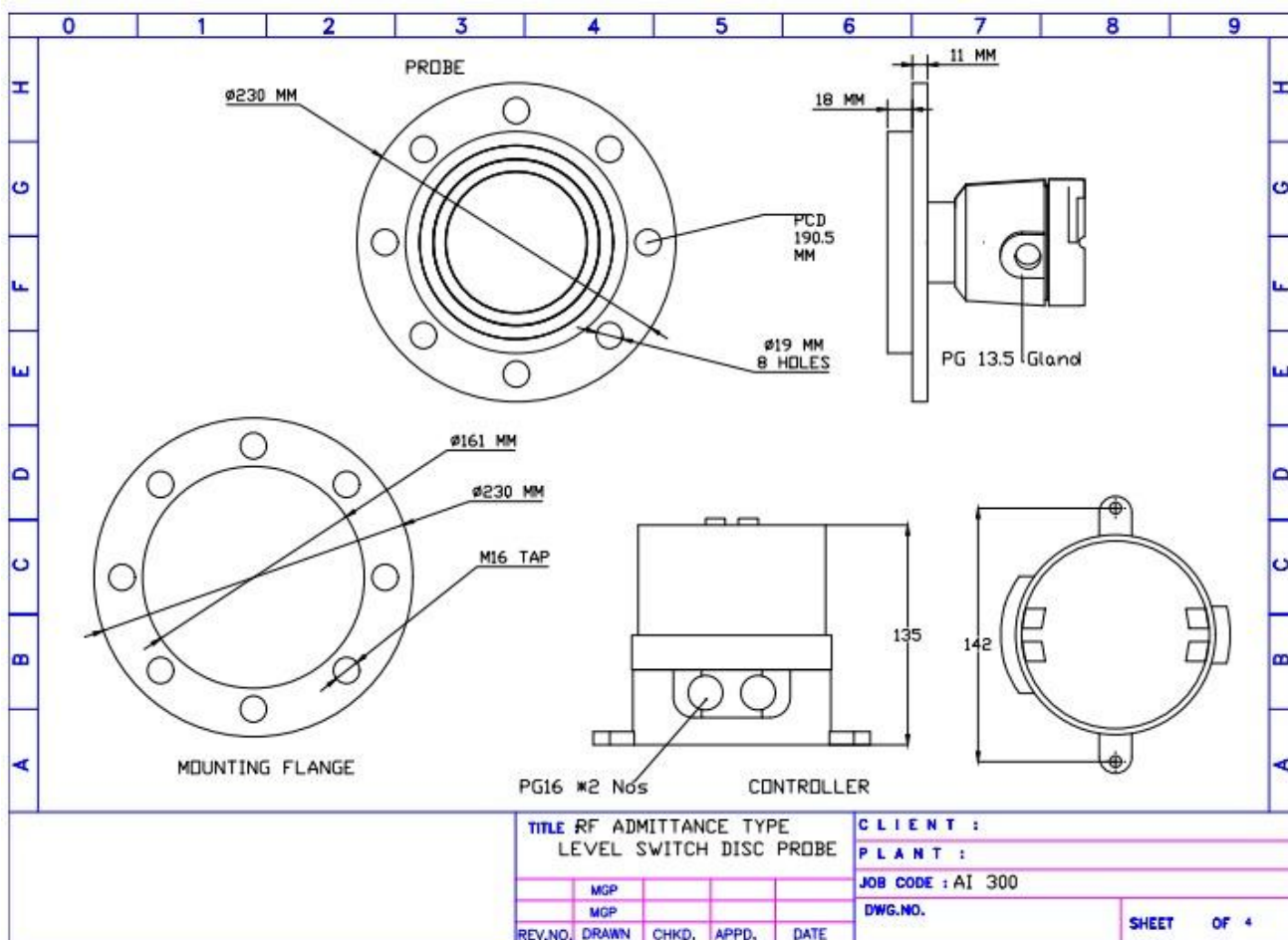
Selection	Code	Description
Temperature	T1/ T2/TS	Max 80°C / 200°/ Customer Specified Special Designed
High Temperature stand off	H8/HS	80MM/Special standoff Extension
Remote Cable	05 / 10 / 15 / 20 / 25 / 30/SR	5 / 10 / 15 / 20 / 25 / 30 m/ Special Customer Requirement
Cable Gland	G1/ G2 / G3/ G4	PG13.5 / ½"NPT / ¾" ET/ PG16
Material Certificate	M0 / MC	Not Required / Material test certificate
Test Certificate	TC / CC	Test Certificate / Calibration Certificate
Documentation	DS / GA / IR	Datasheet / GA / Inspection Report
Tagging	TAG	Customer Tag No.
Probe Length	xxxx	18 mm Fixed

MECHANICAL DRAWING INTEGRAL TYPE :



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MECHANICAL DRAWING REMOTE TYPE :



Technical Specifications:

Before to Order, please confirm below Details,

- Process Parameters: Temperature, pressure,
- atmosphere: hazardous /non-hazardous.

Application media, characteristics: Sp. Gr., Dielectric constant, hazardous or not, corrosiveness, viscosity

Manufactured By:

Levtron Instruments Pvt Ltd.

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